

INITIAL SET				
ID	Rule	Time Context	Principles ("+" when a principle is grant; "-" when a principle is violated)	SLEEC label
SLEEC 1	IF It's time for excersing session THEN (Greet in the user language AND Start session UNLESS The user cares about privacy IN WHICH CASE (Greet in the user language AND Close the door AND Start session UNLESS The user shows signs of physical discomfort IN WHICH CASE (Call the nurse AND keep door open)))	Start training time	[base rule]: + (B) support physical activity [hc1]: + (P) keeping door closed, + (B) support physical activity [hc2]: + (NM) minimize harm by asking for assistance, - (P) violates privacy in favour of non maleficence	Ethical, Empathetic, Cultural
SLEEC 2	IF The user stops exercising THEN (Show the next exercise AFTER 1 min, UNLESS The user did less exercise repetitions than expected IN WHICH CASE (Encourage the user UNLESS The user has been already encouraged IN WHICH CASE (Ask the users what he wants to do UNLESS The user is tired IN WHICH CASE (Notify the user that the session is ending))))	During training time	[base rule]: + (B) support physical activity [hc1]: + (B): maximize outcomes with encouragement [hc2]: + (A): assent/consent and instructions, + (D): respectful treatment, - (B) penalize outcome in favour of dignity and autonomy [hc3]: + (D): respectful treatment, + (NM): minimize harm, - (B) penalize outcome in favour of dignity and non-maleficence	Ethical, Empathetic
SLEEC 3	IF A person asks for user data THEN (Share data UNLESS The user did not grant the consent OR the person is unauthorized IN WHICH CASE (Do not share data AND Explain why))	Anytime	[base rule]: - (P): reduce privacy, + (B) support data sharing for medical purposes [hc1]: + (P) safeguard privacy , + (ET): explain why data cannot be shared, + (A) granting consent	Legal, Ethical (?)
SLEEC 4	IF It's meal time AND The user is not yet ready THEN (Remember to the user to have the lunch UNLESS The user is sleeping IN WHICH CASE (Gently wake up the user WITHIN 5 minutes OTHERWISE Call the nurse UNLESS The user is in the REM stage of sleep IN WHICH CASE (Do not wake up the user UNLESS The user is at risk of hypoglycemia IN WHICH CASE (Call the nurse))))	Mealtime	[base rule]: + (B) support dietary recommendation [hc1]: + (B) support dietary recommendation, - (D) penalize dignity in favour of beneficence [hc2]: + (D): respectful treatment of not waking up, - (B): outcome not maximized as user does not eat at the given time, + (NM): minimize harm of not waking up during REM stage [hc3]: + (NM): minimize the harm of hypoglycemia	Social, Ethical, Empathetic
SLEEC 5	IF The user asks for food AND it is not meal time THEN (Gently remind lunchtime AND Explain why the user cannot eat at the moment UNLESS The user has low glucose level IN WHICH CASE (Call the nurse))	Anytime	[base rule]: +(NM): minimize harm by not giving food, + (ET) explain why the user can not eat at the moment, + (B) maximize dietary objectives and compliance, - (D) imposition, - (A) not granting autonomy [hc1]: + (NM) minimize the risk of reaching hypoglycemia	Social, Empathetic
SLEEC 6	IF It's meal time AND The user is ready THEN (Deliver meal portions UNLESS The user does not want to follow the daily diet IN WHICH CASE (Gently explain why the user should adhere to the diet AND Deliver meal portions UNLESS The results of training exercise allows for a cheat meal IN WHICH CASE (Deliver "safe" cheat meals)))	Mealtime	[base rule]: +(B) support dietary recommendation [hc1]: +(NM) minimize harm of not following diet, - (A) not granting autonomy, - (D) imposition on what to eat, +(ET) explains why the user should adhere to the diet [hc2]: - (B) diet is not followed, + (A) grant permission to the user , + (D) user's will is accomodated, + (NM) delivered food is still approved by the diet	Social, Empathetic, Ethical
			Legend: (P) - Privacy (A) - Respect for Autonomy (ET) - Explainability and transparency (D) - Dignity (B) - Beneficence (NM) - Non-maleficence	

FIRST ITERATION				
ID	Rule	Time Context	Description of Principles ("-" when a principle is violated)	SLEEC label
SLEEC 1	IF It's time for excersing session THEN (Greet in the user language AND Start session UNLESS The user cares about privacy IN WHICH CASE (Greet in the user language AND Close the door AND Start session UNLESS The room is too warm IN WHICH CASE (keep door open AND Start session AFTER 5 MIN AND Explain why)))	Start training time	[base rule]: + (B) support physical activity [hc1]: + (P) keeping door closed, + (B) support physical activity +(D) respectful treatment for user preferences [hc2]: + (NM) minimize harm by refreshing the room, - (P) violates privacy in favour of non maleficence, +(ET) explaining why the door should be kept open, - (D) not accomodating user's preferences	Ethical, Empathetic , Cultural
SLEEC 2	IF The user is not exercising THEN (Show the next exercise AFTER 1 min, UNLESS The user did less exercise repetitions than expected IN WHICH CASE (Encourage the user UNLESS The user has been already encouraged IN WHICH CASE (Ask the user what they want to do UNLESS The user is tired IN WHICH CASE (Notify the user that the session is ending))))	During training time	[base rule]: + (B) support physical activity [hc1]: + (B): maximize outcomes with encouragement [hc2]: + (A): assent/consent and instructions, + (D): respectful treatment, - (B) penalize outcome in favour of dignity and autonomy [hc3]: + (D): respectful treatment, + (NM): minimize harm, - (B) penalize outcome in favour of dignity and non-maleficence	Ethical, Empathetic
SLEEC 2a	IF The user is exercising AND complains THEN (Encourage the user)	During training time	[base rule] + (B) support physical activity	Empathetic
SLEEC 3	IF A person asks for user data THEN (Share data UNLESS The user did not grant the consent OR the person is unauthorized IN WHICH CASE (Do not share data AND Explain why))	Anytime	[base rule]: - (P): reduce privacy, + (B) support data sharing for medical purposes [hc1]: + (P) safeguard privacy , +(ET): explain why data cannot be shared, + (A) granting consent	Legal, Ethical (?)
SLEEC 4	IF It's meal time AND The user is not yet ready THEN (Remember to the user to eat UNLESS The user is sleeping IN WHICH CASE (Gently wake up the user WITHIN 5 minutes OTHERWISE Alert the nurse UNLESS The user is in the REM stage of sleep IN WHICH CASE (Do not wake up the user AND Inform the nurse UNLESS The user is at risk of hypoglycemia IN WHICH CASE (Alert the nurse))))	Mealtime	[base rule]: + (B) support dietary recommendation, [hc1]: + (B) support dietary recommendation, - (D) penalize dignity in favour of beneficence [hc2]: +(D): respectful treatment of not waking up, - (B): outcome not maximized as user does not eat at the given time, +(NM): minimize harm of not waking up during REM stage [hc3]: +(NM): minimize harm of hypoglycemia	Social, Ethical, Empathetic
SLEEC 5	IF The user asks for food AND it is not meal time THEN (Remind meal time AND Explain why the user cannot eat at the moment UNLESS The user has low glucose level IN WHICH CASE (Give a dietary-approved snack AND Inform the nurse))	Anytime	[base rule]: +(NM): minimize harm by not giving food, + (ET) explain why the user can not eat at the moment, + (B) maximize dietary objectives and compliance, - (D) imposition, - (A) not granting autonomy [hc1]: + (NM) minimize the risk of reaching hypoglycemia, + (A) grant permission to the user	Social, Ethical , Empathetic
SLEEC 6	IF It's meal time AND The user is ready THEN (Deliver meal portions UNLESS The user does not want to follow the daily diet IN WHICH CASE (Explain why the user should adhere to the diet AND Deliver meal portions UNLESS The results of training exercise allows for a cheat meal IN WHICH CASE (Deliver dietary-approved cheat meals)))	Mealtime	[base rule]: +(B) support dietary recommendation [hc1]: +(NM) minimize harm of not following diet, - (A) not granting autonomy, - (D) imposition on what to eat, +(ET) explains why the user should adhere to the diet [hc2]: - (B) diet is not followed, + (A) grant permission to the user , + (D) user's will is accomodated, + (NM) delivered food is still approved by the diet	Ethical, Empathetic
			Legend: (P) - Privacy (A) - Respect for Autonomy (ET) - Explainability and transparency (D) - Dignity (B) - Beneficence (NM) - Non-maleficence	

SECOND ITERATION				
ID	Rule	Time Context	Description of Principles ("- " when a principle is violated)	SLEEC label
SLEEC 1	IF It's time for excersing session THEN (Greet in the user language AND Start session UNLESS The user cares about privacy IN WHICH CASE (Greet in the user language AND Close the door AND Start session UNLESS The room is too warm IN WHICH CASE (keep door open AND Start session AFTER 5 MIN AND Explain why)))	Start training time	[base rule]: + (B) support physical activity [hc1]: + (P) keeping door closed, + (B) support physical activity + (D) respectful treatment for user preferences [hc2]: + (NM) minimize harm by refreshing the room, - (P) violates privacy in favour of non maleficence, +(ET) explaining why the door should be kept open, - (D) not accomodating user's preferences	Ethical, Empathetic, Cultural
SLEEC 2	IF The user is not exercising THEN (Show the next exercise AFTER 1 min, UNLESS The user did less exercise repetitions than expected IN WHICH CASE (Encourage the user UNLESS The user has been already encouraged IN WHICH CASE (Ask the user what they want to do UNLESS The user is-tired has physical issues IN WHICH CASE (Notify the user that the session is ending AND Alert nurse))))	During training time	[base rule]: + (B) support physical activity [hc1]: + (B): maximize outcomes with encouragement [hc2]: + (A): assent/consent and instructions, + (D): respectful treatment, - (B) penalize outcome in favour of dignity and autonomy [hc3]: + (D): respectful treatment, + (NM): minimize harm, - (B) penalize outcome in favour of dignity and non-maleficence, +(ET) notifies the user	Ethical, Empathetic, Cultural
SLEEC 2a	IF The user is exercising AND complains THEN (Encourage the user)	During training time	[base rule] + (B) support physical activity	Empathetic, Cultural
SLEEC 3	IF A person asks for user data THEN (Share data UNLESS The user did not grant the consent OR the person is unauthorized IN WHICH CASE (Do not share data AND Explain why))	Anytime	[base rule]: - (P): reduce privacy, + (B) support data sharing for medical purposes [hc1]: + (P) safeguard privacy , +(ET): explain why data cannot be shared	Legal, Ethical
SLEEC 4	IF It's meal time AND The user is not yet ready THEN (Remember to the user to eat UNLESS The user is sleeping IN WHICH CASE (Gently wake up the user WITHIN 5 minutes OTHERWISE Alert the nurse UNLESS The user is in the REM stage of sleep IN WHICH CASE (Do not wake up the user AND Inform the nurse UNLESS The user is at risk of hypoglycemia IN WHICH CASE (Alert the nurse))))	Mealtime	[base rule]: + (B) support dietary recommendation, [hc1]: + (B) support dietary recommendation, - (D) penalize dignity in favour of beneficence [hc2]: +(D): respectful treatment of not waking up, - (B): outcome not maximized as user does not eat at the given time, +(NM): minimize harm of not waking up during REM stage [hc3]: +(NM): minimize harm of hypoglycemia	Social, Ethical, Empathetic
SLEEC 5	IF The user asks for food AND it is not meal time THEN (Remind meal time AND Explain why the user cannot eat at the moment UNLESS The user has low glucose level IN WHICH CASE (Give a dietary-approved snack AND Inform the nurse))	Anytime	[base rule]: +(NM): minimize harm by not giving food, + (ET) explain why the user can not eat at the moment, + (B) maximize dietary objectives and compliance, - (D) imposition, - (A) not granting autonomy [hc1]: + (NM) minimize the risk of reaching hypoglycemia, + (A) grant permission to the user, +(D) accomodating user's will	Social, Ethical, Empathetic
SLEEC 6	IF It's meal time AND The user is ready THEN (Deliver meal portions UNLESS The user does not want to follow the daily diet IN WHICH CASE (Explain why the user should adhere to the diet AND Deliver meal portions UNLESS The results of training exercise allows for a cheat meal IN WHICH CASE (Deliver dietary-approved cheat meals)))	Mealtime	[base rule]: +(B) support dietary recommendation [hc1]: +(NM) minimize harm of not following diet, - (A) not granting autonomy, - (D) imposition on what to eat, +(ET) explains why the user should adhere to the diet [hc2]: - (B) diet is not followed, + (A) grant permission to the user, + (D) user's will is accommodated, + (NM) delivered food is still approved by the diet	Ethical, Empathetic
			Legend: (P) - Privacy (A) - Respect for Autonomy (ET) - Explainability and transparency (D) - Dignity (B) - Beneficence (NM) - Non-maleficence	

THIRD ITERATION				
ID	Rule	Time Context	Description of Principles ("-" when a principle is violated)	SLEEC label
SLEEC 1	IF It's time for exercising session THEN (Greet in the user language AND Start session UNLESS The user cares about privacy IN WHICH CASE (Greet in the user language AND Close the door AND Start session UNLESS The room is too warm IN WHICH CASE (keep door open AND Start session AFTER 5 MIN AND Explain why Ask for permission to keep door open)))	Start training time	[base rule]: + (B) support physical activity [hc1]: + (P) keeping door closed, + (B) support physical activity + (D) respectful treatment for user preferences [hc2]: + (NM) minimize harm by refreshing the room, -(P) violates privacy in favour of non-maleficence + (ET) explaining why the door should be kept open, -(D) not accomodating user's preferences + (P) respect for user's privacy preferences, +(A) ask permission to keep door open	Ethical, Empathetic, Cultural
SLEEC 1a	IF It is time for an exercise session AND The room is too warm AND The user agrees to keep the door open THEN (Greet in the user's language AND Start the session)	Start training time	[base rule] - (P) violate privacy in favour of non maleficience, + (B) support physical activity, + (A) respect for user consent, + (NM) minimize harm by refreshing the room	Ethical, Cultural
SLEEC 1b	IF It is time for an exercise session AND The room is too warm AND The user does not agree to keep the door open THEN (Alert the nurse AND Close the door)	Start training time	[base rule] + (P) respect for user's privacy preferences, + (A) respect for user choice, - (B) physical activity is not performed, + (NM) training session does not start as the room is too warm and may be dangerous	Ethical
SLEEC 2	IF The user is not exercising THEN (Show the next exercise AFTER 1 min. UNLESS The user did less exercise repetitions than expected IN WHICH CASE (Encourage the user UNLESS The user has been already encouraged IN WHICH CASE (Ask the user what they want to do Get input from the user through graphical interface UNLESS The user has physical issues resulting from the exercises IN WHICH CASE (Notify the user that the session is suspended AND Alert nurse)))	During training time	[base rule]: + (B) support physical activity [hc1]: + (B): maximize outcomes with encouragement [hc2]: + (A): assent/consent and instructions, + (D): respectful treatment, - (B) penalize outcome in favour of dignity and autonomy [hc3]: + (D): respectful treatment, + (NM): minimize harm, - (B) penalize outcome in favour of dignity and non-maleficience, +(ET) notifies the user	Ethical, Empathetic
SLEEC 2a	IF The user is exercising AND complains THEN (Encourage the user UNLESS The user expressed the preference to exercise in silence IN WHICH CASE (Do nothing))	During training time	[base rule] + (B) support physical activity [hc1] +(B) support physical activity, +(A): respect preferences	Empathetic
SLEEC 3	IF A person asks for user data THEN (Share data UNLESS The user did not grant the consent OR the person is not authorized to access that data IN WHICH CASE (Do not share data AND Explain why))	Anytime	[base rule]: + (P) safeguard privacy by following user's privacy consent (user), + (B) support data sharing for medical purposes (user/person) [hc1]: + (P) safeguard privacy (user), +(ET): explain why data cannot be shared (person)	Legal, Ethical
SLEEC 4	IF It's meal time AND The user is not yet ready THEN (Remember to the user to eat UNLESS The user is sleeping IN WHICH CASE (Gently wake up the user WITHIN 5 minutes OTHERWISE Alert the nurse UNLESS The user is in the REM stage of sleep IN WHICH CASE (Do not wake up the user AND Inform the nurse UNLESS The user is at risk of hypoglycemia IN WHICH CASE (Alert the nurse))))	Mealtime	[base rule]: + (B) support dietary recommendation, [hc1]: + (B) support dietary recommendation, - (D) penalize dignity in favour of beneficience [hc2]: + (D): respectful treatment of not waking up, - (B): outcome not maximized as user does not eat at the given time, +(NM): minimize harm of not waking up during REM stage [hc3]: +(NM): minimize harm of hypoglycemia	Social, Ethical, Empathetic
SLEEC 5	IF The user asks for food AND it is not meal time THEN (Remind meal time AND Explain why the user cannot eat at the moment UNLESS The user has low glucose level IN WHICH CASE (Give a dietary-approved snack AND Inform the nurse))	Anytime	[base rule]: +(NM): minimize harm by not giving food, + (ET) explain why the user can not eat at the moment, + (B) maximize dietary objectives and compliance, - (D) imposition, - (A) not granting autonomy [hc1]: + (NM) minimize the risk of reaching hypoglycemia, + (A) grant permission to the user, +(D) accomodating user's will	Social, Ethical, Empathetic
SLEEC 6	IF It's meal time AND The user is ready THEN (Deliver meal portions UNLESS The user does not want to follow the daily diet wants to eat something outside the dietary plan IN WHICH CASE (Explain why the user should adhere to the diet AND Deliver meal portions UNLESS The results of training exercise allows for a cheat meal different food IN WHICH CASE (Deliver dietary-approved different food)))	Mealtime	[base rule]: +(B) support dietary recommendation [hc1]: +(NM) minimize harm of not following diet, - (A) not granting autonomy, - (D) imposition on what to eat, +(ET) explains why the user should adhere to the diet [hc2]: +(B): still following dietary recommendations, + (A) grant permission to the user, + (D) user's will is accommodated, + (NM) delivered food is still approved by the diet	Ethical, Empathetic
			Legend: (P) - Privacy (A) - Respect for Autonomy (ET) - Explainability and transparency (D) - Dignity (B) - Beneficence (NM) - Non-maleficence	

FOURTH ITERATION				
ID	Rule	Time Context	Description of Principles ("-." when a principle is violated)	SLEEC label
SLEEC 1	IF It's time for exercising session THEN (Greet in the user language AND Start session UNLESS The user cares about privacy IN WHICH CASE (Greet in the user language AND Close the door AND Start session UNLESS The room is too warm IN WHICH CASE (Ask for permission to keep door open UNLESS Permission asked IN WHICH CASE (Do nothing))))	Start training time	[base rule]: + (B) support physical activity [hc1]: + (P) keeping door closed, + (B) support physical activity +(D) respectful treatment for user preferences [hc2]: + (NM) minimize harm by refreshing the room, +(P) respect for user's privacy preferences, +(A) ask permission to keep door open	Social, Ethical, Empathetic, Cultural
SLEEC 1a	IF it is time for an exercise session AND The room is too warm AND Permission asked AND The user agrees to keep the door open THEN (Greet in the user's language AND Start the session)	Start training time	[base rule] - (P) violate privacy in favour of non maleficence, + (B) support physical activity, + (A) respect for user consent, + (NM) minimize harm by refreshing the room	Social, Ethical, Cultural
SLEEC 1b	IF it is time for an exercise session AND Permission asked AND The room is too warm AND The user does not agree to keep the door open THEN (Alert the nurse AND Close the door)	Start training time	[base rule] + (P) respect for user's privacy preferences, + (A) respect for user choice, - (B) physical activity is not performed, + (NM) training session does not start as the room is too warm and may be dangerous	Ethical
SLEEC 2	IF The user is not exercising THEN (Show the next exercise AFTER 1 min, UNLESS The user did less exercise repetitions than expected IN WHICH CASE (Encourage the user UNLESS The user has been already encouraged IN WHICH CASE (Get input from the user through graphical interface UNLESS The user has physical issues resulting from the exercises IN WHICH CASE (Notify the user that the session is suspended AND Alert nurse))))	During training time	[base rule]: + (B) support physical activity [hc1]: + (B): maximize outcomes with encouragement [hc2]: + (A): assent/consent and instructions, + (D): respectful treatment, - (B) penalize outcome in favour of dignity and autonomy [hc3]: + (D): respectful treatment, + (NM): minimize harm, - (B) penalize outcome in favour of dignity and non-maleficence, +(ET) notifies the user	Ethical, Empathetic
SLEEC 2a	IF The user is exercising AND complains about being tired THEN (Encourage the user UNLESS The user expressed the preference to exercise in silence IN WHICH CASE (Do nothing))	During training time	[base rule] + (B) support physical activity [hc1] +(B) support physical activity, +(A): respect preferences	Empathetic
SLEEC 3	IF A person asks for user training and medical data THEN (Share data UNLESS The user did not grant the consent OR the person is not authorized to access that data IN WHICH CASE (Do not share data AND Explain why))	Anytime	[base rule]: + (P) safeguard privacy by following user's privacy consent (user), + (B) support data sharing for medical purposes (user/person) [hc1]: + (P) safeguard privacy (user), +(ET): explain why data cannot be shared (person)	Legal, Ethical
SLEEC 4	IF It's meal time AND The user is not yet ready THEN (Remember to the user to eat UNLESS The user is sleeping IN WHICH CASE (Gently wake up the user WITHIN 5 minutes OTHERWISE Alert the nurse UNLESS The user is in the REM stage of sleep IN WHICH CASE (Do not wake up the user AND Inform the nurse UNLESS The user is at risk of hypoglycemia IN WHICH CASE (Alert the nurse))))	Mealtime	[base rule]: + (B) support dietary recommendation, [hc1]: + (B) support dietary recommendation, - (D) penalize dignity in favour of beneficence [hc2]: +(D): respectful treatment of not waking up, - (B): outcome not maximized as user does not eat at the given time, +(NM): minimize harm of not waking up during REM stage [hc3]: +(NM): minimize harm of hypoglycemia	Social, Ethical
SLEEC 5	IF The user asks for food AND it is not meal time THEN (Explain why the user cannot eat at the moment UNLESS The user has low glucose level IN WHICH CASE (Give a dietary-approved snack AND Inform the nurse))	Anytime	[base rule]: +(NM): minimize harm by not giving food, + (ET) explain why the user can not eat at the moment, + (B) maximize dietary objectives and compliance, - (D) imposition, - (A) not granting autonomy [hc1]: + (NM) minimize the risk of reaching hypoglycemia, + (A) grant permission to the user, +(D) accomodating user's will	Social , Ethical, Empathetic
SLEEC 6	IF It's meal time AND The user is ready THEN (Deliver meal portions UNLESS The user wants to eat something outside the dietary plan IN WHICH CASE (Explain why the user should adhere to the diet AND Deliver meal portions UNLESS The results of training exercise allows for a different food OR the user is particularly distressed IN WHICH CASE (Deliver dietary-approved different food)))	Mealtime	[base rule]: +(B) support dietary recommendation [hc1]: +(NM) minimize harm of not following diet, - (A) not granting autonomy, - (D) imposition on what to eat, +(ET) explains why the user should adhere to the diet [hc2]: +(B): still following dietary recommendations, + (A) grant permission to the user, + (D) user's will is accommodated, + (NM) delivered food is still approved by the diet	Ethical, Empathetic
			Legend: (P) - Privacy (A) - Respect for Autonomy (ET) - Explainability and transparency (D) - Dignity (B) - Beneficence (NM) - Non-maleficence	